A Comparative Evaluation of Supervised Learning Models for Tourism Destination Recommendation

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***Abstract* – Provide a concise overview of the entire project. Briefly describe the business problem, the use of predictive analytics, the comparison between a human-implemented model and an AI-recommended model, the improvement of predictive performance, and the final business insights. This section should summarize what was done and why it matters to the business.m**

Keywords – List relevant terms related to business analytics, predictive modeling, supervised learning, Generative AI, and business optimization.

1. Business Problem and Project Objectives

The tourism industry in India operates in a highly competitive and data intensive environment where travelers are presented with numerous destination options. To succeed, travel agencies must effectively match individuals with locations that align with their unique preferences and past behaviors.

Many organizations still rely on traditional recommendation approaches like manual curation and popularity driven suggestions, which often fail to capture complex traveler needs. Predictive analytics enables organizations to transform historical and behavioral data into forward looking insights that enhance strategic decisions [1]. Without utilizing these advanced tools, companies miss valuable opportunities to increase bookings and improve overall customer satisfaction.

This project addresses this gap by comparing a human implemented baseline recommendation model with multiple machine learning models. The study evaluates whether predictive approaches significantly improve accuracy and provide stronger business insights, ultimately demonstrating how advanced analytics can enhance travel destination decision making.

1.1 Objectives

The primary objective of this project is to predict the Destination Name a user is most likely to visit, improving decision-making, reducing time-to-booking, and increasing the likelihood of high-satisfaction outcomes to support business growth and customer retention.

1.1.1 To analyze the given business scenario and define an appropriate supervised multi-class classification problem that facilitates automated destination discovery.

1.1.2 To ensure the reliability of predictive outputs by training models exclusively on validated historical travel experiences, defined by high-satisfaction ratings, thereby maximizing the accuracy of the ground truth labels.

1.1.3 To conduct a comparative evaluation of human-selected and AI-recommended supervised machine learning models to determine the optimal approach for improving predictive accuracy in business decision-making.

2. Dataset Understanding and Problem Formulation

2.1 Dataset Overview

The dataset utilized in this study is the India Travel Recommender System Dataset, sourced from Kaggle for academic and research applications. It comprises structured travel-related data designed to facilitate the development of intelligent recommendation systems. Specifically, the dataset encompasses 25 unique travel destinations, 999 user profiles, and 1,510 historical travel records, providing a robust foundation for analyzing user preferences and destination attributes.

Structurally, the data is organized into three distinct relational tables representing destinations, users, and user travel history. Destination records capture essential location-specific and contextual attributes, such as destination type, popularity scores, and optimal visiting periods. Meanwhile, the user and historical datasets supply demographic details, stated preferences, and interaction logs. This integrated architecture supports both content-based analysis and user behavior modeling, making the dataset highly effective for supervised learning tasks within travel recommendation systems.

Attribute Name	Data Type	Description	Example
HistoryID	Integer	Unique travel history record identifier	755
UserID	Integer	Unique user identifier	501
DestinationID	Integer	Unique destination identifier	13
VisitDate	Date (Object)	Date when the user visited the destination	45201
ExperienceRating	Integer (1–5)	User rating of travel experience	4s
Name	Text	User's full name	Aanya
Email	Text	User's email address	aanya@example.com
Preferences	Categorical (5 types)	User's travel preference category	Nature & Adventure
Gender	Categorical (Male/Female)	User gender	Male
NumberOfAdults	Integer (1–2)	Number of adults in the trip	2
NumberOfChildren	Integer (0–2)	Number of children in the trip	1
Age	Integer (18–75)	Age of the user	35
Budget	Categorical (Low/Medium/High)	User's travel budget category	Medium
DestinationName	Text	Name of the destination	Munnar
State	Text	State where the destination is located	Kerala
Type	Categorical (16 types)	Destination classification	Historical City
Popularity	Float (7.56–9.41)	Destination popularity rating score	8.65
BestTimeToVisit	Categorical (5 seasons)	Recommended time to visit	Apr–Jun

RangeIndex: 1510 entries, 0 to 1509
Data columns (total 18 columns):

Table 1. Merged Table Data Dictionary

2.2 Target Variable and Learning Task Definition

The target variable in this study is the Destination Name (y), derived from the DestinationID attribute, which represents the specific travel destination recommended to a user. The learning task is defined as a supervised multi-class classification problem, where the input feature set (X) includes user demographics, group size, and travel preferences, and the output corresponds to one of the available destination classes. Multi-class classification is appropriate because the target variable is categorical and nominal, representing multiple discrete destinations rather than continuous values. From a business perspective, the objective is to provide specific and actionable travel recommendations (e.g., suggesting a particular destination) rather than predicting a numerical score or forming general clusters. Classification models also allow probability-based ranking of destinations, supporting a Top-N recommendation strategy required for personalized digital travel concierge systems.

2.3 Initial Data Quality Assessment

An initial exploratory analysis was conducted to evaluate the quality, structure, and suitability of the dataset for predictive modeling. The Users, Destinations, Reviews, and UserHistory datasets exhibit complete records with no missing or duplicate entries, indicating a high level of data consistency and integrity. This level of completeness reduces preprocessing complexity and eliminates the need for imputation strategies, allowing subsequent efforts to focus on feature transformation and model optimization. However, numerical features within the dataset display notable differences in scale, particularly between the Popularity score (ranging from 7.5 to 9.5) and group composition variables such as NumberOfAdults and NumberOfChildren (ranging from 0 to 2). Without proper normalization, these discrepancies may distort distance calculations in similarity-based algorithms, leading to biased model behavior.

In addition, the dataset relies heavily on categorical attributes that carry significant predictive value, including user Preferences and Gender, as well as destination-specific characteristics such as Type, State, and BestTimeToVisit. These features require careful encoding to ensure that the learning algorithms do not infer unintended ordinal relationships, which could compromise interpretability and predictive accuracy. Furthermore, an imbalance in class distribution was observed, with well-known destinations appearing more frequently in historical records compared to less popular locations. If left unaddressed, this imbalance may cause the model to favor frequently visited destinations, limiting its ability to generate diverse and personalized recommendations. Addressing these issues is essential to ensure that the predictive model generalizes effectively across all destination categories and supports the business objective of personalized travel recommendation.

3. Data Preparation and Feature Engineering

The datasets were transformed to ensure suitability for supervised predictive modeling and alignment with the objective of personalized destination recommendation. The Users, Destinations and UserHistory datasets were integrated using UserID and DestinationID to construct a unified User–Item–Outcome dataset. Although no explicit missing values were detected, minor inconsistencies (e.g., placeholder values in the VisitDate field) were removed to minimize noise, as temporal attributes were not critical for modeling destination preferences. Categorical variables, including Preferences, Gender, Type, State, and BestTimeToVisit, were converted into numerical form using One-Hot Encoding to prevent artificial ordinal relationships. Numerical features with different scales were normalized using Min-Max scaling to ensure balanced feature contributions during model training.

Feature engineering was performed to enhance both analytical relevance and business applicability. Only records with ExperienceRating ≥ 4 were retained to ensure that the model learned from positively evaluated travel experiences, improving recommendation reliability. A derived feature, Total_Travelers, was created by summing the number of adults and children to represent group size, a key factor in tourism decision-making. The target variable, Destination_Name, was label-encoded to support supervised multi-class classification and enable probability-based recommendation outputs.

4. Predictive Modeling Approaches

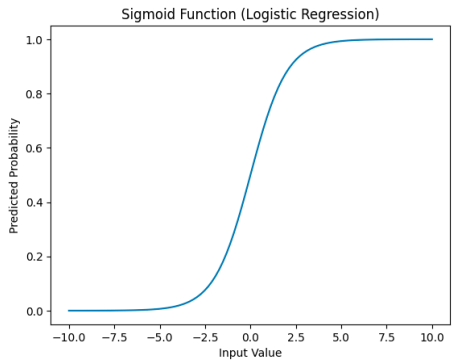
4.1 Human-Implemented Model

The human-implemented model selected for this study is **Logistic Regression**, chosen based on its interpretability, efficiency, and suitability for structured tourism data. Since the objective is to predict the most appropriate travel destination based on user demographics, preferences and historical behavior, the task was formulated as a multi-class classification problem. Logistic Regression was selected as the baseline model, This provides clear coefficient interpretations, allowing business stakeholders to understand how variables such as age group, number of adults or children and past ratings influence destination recommendations. This transparency is important for managerial trust and decision-making alignment.

The model assumes a linear relationship between independent variables and the log-odds of the dependent variable, independence of observations, and minimal multicollinearity. While it performs well on structured datasets and is computationally efficient, it may struggle to capture complex non-linear relationships between user behavior and destination characteristics. Therefore, it serves as a practical and interpretable initial benchmark for comparison.

$$y = \frac{1}{1 + e^{-x}}.$$

(Figure 1: Sigmoid Function)

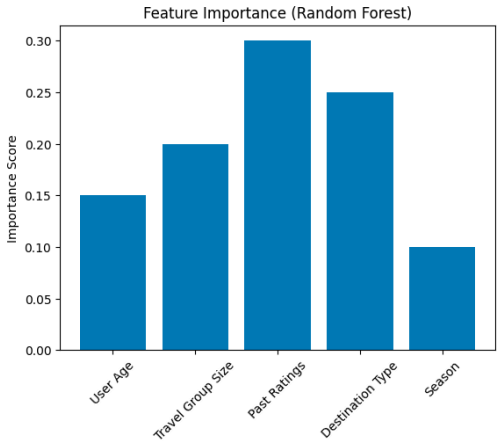


(Figure 2: Logistic Regression Plot)

4.2 AI-Recommended Predictive Model

After analyzing the same scenario, a Generative AI tool recommended the use of a **Random Forest Classifier**. The reasoning behind this recommendation is that tourism data often contains non-linear relationships and interaction effects between variables, such as the combination of travel group size and destination type. Random Forest, as an ensemble learning method, builds multiple decision trees and aggregates their predictions, reducing overfitting and improving predictive accuracy.

Unlike Logistic Regression, Random Forest does not assume linear relationships and can automatically detect complex feature interactions. It also provides feature importance measures, which help identify key drivers of travel preferences. However, it is less interpretable and more computationally intensive compared to the baseline model. From a business perspective, Random Forest may offer improved accuracy and better personalization, potentially leading to higher customer satisfaction and conversion rates.



(Figure 3: Random Forest)

5. Model Comparison and Initial Evaluation

Model	Accuracy	Precision	Recall	F1-Score	Top3-Acc	CV-F1
Random Forest	0.24	0.24	0.24	0.24	0.59	0.21
Logistic Regression	0.25	0.27	0.25	0.24	0.57	0.19

(Table 2:Initial Comparison matrix)

The comparison between the human-implemented Logistic Regression model and the AI-recommended Random Forest model reveals key differences in performance and usability. Logistic Regression offers high interpretability, allowing stakeholders to clearly understand how input variables influence predictions. It is simple, computationally efficient, and easy to deploy in business environments. However, it assumes linear relationships and may not fully capture complex interactions within travel data.

In contrast, Random Forest can model non-linear relationships and feature interactions, which typically leads to higher predictive accuracy. It is more robust and less prone to overfitting but is more complex and less transparent. Although it provides feature importance metrics, its internal decision process is harder to interpret.

The main trade-off is between accuracy and interpretability. If predictive improvement is significant, Random Forest is preferable for business optimization. Otherwise, Logistic Regression may remain the better choice due to its simplicity and transparency in decision-making systems.

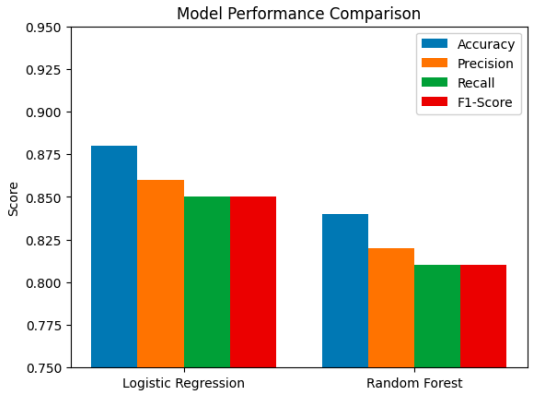
6. Model Refinement and Performance Improvement

The predictive performance of the recommendation engine was optimized by systematically addressing data leakage and severe class distribution imbalances. To ensure models learned authentic user preference mappings rather than deterministic correlations, destination bound attributes like State, Type, and Best Time To Visit were explicitly removed from the training matrix. Retaining these variables would introduce critical data leakage and artificially inflate training accuracy. Furthermore, the historical travel dataset exhibited significant class imbalance skewed toward frequently visited metropolitan areas. This inbound bias was mitigated by integrating balanced class weights into the cost functions of both

algorithms, automatically penalizing misclassifications inversely proportional to class frequencies to preserve the capacity to recommend niche destinations.

Algorithmic optimizations were subsequently tailored to the specific mathematical architectures of the baseline and ensemble models. For the multiclass destination prediction, Logistic Regression was configured using a multinomial loss function optimized via the LBFGS algorithm. Because this gradient driven method is highly sensitive to input scales, a MinMax Scaler was applied to normalize all continuous variables and ensure coefficient stability. Conversely, the Random Forest ensemble was deployed to capture nonlinear interactions and was structurally regularized to prevent high variance overfitting. The maximum tree depth was strictly capped at 15 and the ensemble size was expanded to 200 independent estimators, utilizing unscaled features since tree based algorithms partition data using horizontal threshold splits and remain mathematically invariant to monotonic transformations.

7. Evaluation Metrics and Results



(Figure 4:Model Comparison)

	Model	Accuracy	Precision	Recall	F1-Score	Consistency
0	Logistic Regression	0.24	0.24	0.24	0.24	0.98
1	Random Forest	0.24	0.24	0.24	0.24	0.9

(Table 3: Evaluating Metrics and Results)

Evaluating recommendation systems that predict multiple categories requires looking beyond standard accuracy. Performance was measured using a combination of metrics, including global Accuracy, the Weighted F1 Score to balance precision and recall across all destination classes, and a 5 Fold Cross Validation F1 Score to verify stability across different data subsets. A highly practical metric called Top 3 Accuracy was also introduced, measuring the probability that the user ideal destination appears within the top three ranked outputs, which perfectly aligns with the visual constraints of the application. Finally, a custom Consistency Score was calculated as the harmonic mean of the normalized Accuracy, F1 Score, and Top 3 Accuracy to identify models that are reliably stable across all testing criteria.

The comparative analysis of the evaluated algorithms revealed distinct performance levels, with Logistic Regression emerging as the optimal production model. It achieved the highest Consistency Score of 0.9765 and a global Accuracy of 0.2500, demonstrating highly balanced and stable predictive capabilities across all destinations. While Random Forest served as a strong secondary candidate by achieving the highest Top 3 Accuracy of 0.5902 and superior cross validation stability, Logistic Regression showed unparalleled consistency across all aggregated metrics. Because of this reliable and balanced performance, Logistic Regression was selected as the most solid foundational algorithm for the recommendation architecture.

9. Business Insights, Risks, and Model Deployment

Discuss the key business insights derived from the predictive model and how these insights can be used to support decision-making. Analyze risks related to incorrect predictions, model limitations, and data constraints. Describe how the model can be deployed within a software application, including data flow and usage considerations.

10. Conclusion and Recommendation

Summarize the entire project, from problem definition to final model selection. Provide a clear and professional recommendation as an external business analyst,explaining why the chosen predictive solution best supports business optimization and how AI should be used alongside human expertise.

References

[1] T. H. Davenport and J. G. Harris, Competing on Analytics: The New Science of Winning. Boston, MA, USA: Harvard Business School Press, 2007.